

The Darwinian Mind of the Machine: Rethinking LLM Training as Evolution

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Abstract

This essay challenges the prevailing metaphor of “learning” used to describe Large Language Model (LLM) training, proposing instead that these systems represent a form of hyper-accelerated, data-driven evolution. Through analysis of Daniel Dennett’s hierarchy of evolutionary competence and examination of the poverty of the stimulus problem, we argue that LLMs are Darwinian creatures evolved at computational speeds in environments of pure text. This framework explains their linguistic capabilities through convergent evolution rather than learning, resolves paradoxes about their competence without understanding, and for our understanding of the relevance for these models for generative grammar.

1 Introduction

In the recent discussion on the plausibility of generative grammar, few developments have captured our collective imagination quite like Large Language Models (LLMs). These systems seem to be able to pass tests of linguistic competence that seem to have hitherto been reserved for human beings: not just do they produce remarkably grammatical prose — at least in well-documented languages such as English —, but they also seem to be able to give grammaticality judgements of remarkable subtlety (Hu et al., 2024; Qiu et al., 2024; Mulders and Ruys, 2024).

Faced with such remarkable capabilities, people have instinctively reached for familiar metaphors to make sense of this behavior which, if we are honest, were completely unexpected even as late as five years ago. We speak of machines that “learn”, systems that “understand”, and models that “know” things about the world. This anthropomorphic framing, while psychologically comforting, represents a profound category error that obscures the true nature of these systems. When we describe an LLM as “learning” language, we inadvertently import assumptions about consciousness, intentionality, and experience that simply do not apply.

More to the point it has led scholars such as Piantadosi (2024) and Hinton, in several interviews (see for instance the phone interview with him on being awarded the Nobel prize for Physics, <https://www.nobelprize.org/prizes/physics/2024/hinton/interview/>) to claim that LLMs are the death blow to the generative enterprise or any idea of an innate Universal Grammar. If LLMs can acquire language without any predetermined setting, this at least forms a proof of existence for the claim that humans can also learn language as a blank slate. This has sparked a number of backs and forths between generative grammarians, who maintain that LLMs have little to do with the human Faculty of language (see for instance Murphy et al. 2025, and scholars who maintain that LLMs show that the assumptions of Generative Grammar are definitively proven wrong (Piantadosi and Yang, 2022), with some scholars trying to move beyond these contrasts and to find a new path for linguistics (see Müller 2025 and more recently Moro 2025).

But what if this entire conceptual framework is not merely inadequate, but actively misleading? What if the process that creates LLMs bears no resemblance to learning as we understand it, and instead represents something rather different? We propose we abandon the metaphor of learning entirely and adopt a more accurate framework: LLM training is a form of hyper-accelerated, data-driven evolution—a Darwinian process operating at computational speeds that compress thousands of years of natural selection into months of GPU computation. This shift in perspective does more than provide a clearer technical understanding; it fundamentally reframes our relationship with these

systems and offers profound insights into the nature of natural language itself.

2 The Learning Paradox and the Poverty of the Stimulus

Central to the discussion is an old problem in cognitive science: the poverty of the stimulus. This argument, most forcefully articulated by linguist Noam Chomsky (Chomsky, 1986), observes that human language acquisition presents a seemingly impossible feat. Children master the intricate rules of (at least) the syntax of their native language despite being exposed to linguistic data that is both limited and often imperfect.

Consider a concrete example from Dutch. The phrase ‘dat is’ (*that is* can be contracted to ‘das’ (‘dat is goed’ (*that is good* can be pronounced approximately as ‘das goed’, but not after *wh*-movement: ‘ik weet wat dat is’ (*I know what that is* cannot be contracted to ‘ik weet wat das’. Contraction is not taught to children – if anything people advise against it as being too informal across the board: no parent explicitly teaches their child the difference between sentences with and without *wh*-extraction. Yet children intuitively grasp this distinction, along with thousands of similar grammatical subtleties in all languages that have been studied in sufficient detail. The input the children receive contains insufficient evidence to deduce these rules through pure statistical analysis.¹

Observations like this led Chomsky to propose the existence of a Universal Grammar (UG): an innate biological endowment that structures language acquisition. According to this theory, humans are born with a genetic blueprint for language that allows us to extract grammatical principles from impoverished data. While the idea of a UG has remained invaried through the years, what it "contains" has changed radically: from a whole set of parameterized principles, as well as an X-bar innate structure (Chomsky, 1970), postulated in Chomsky (1981) and subsequent work, to the bare existence of an operation Merge (Chomsky, 1995) and the parametrization shifted to lexical items, and

¹Or at least this has been the idea for a long time. In an informal test, it turns out that Google Gemini can distinguish between the two sentences, calling the second one more unusual ‘because it does not appear at the beginning of the embedded clause’. (Session with Google Gemini 2.5Pro on September 8, 2025.)

functional categories in particular, known as the Borer-Chomsky Conjecture (named such by Baker 2008).

Whichever definition we consider of UG, LLMs at first blush seem to present a challenge to this line of thinking. These systems, too, are exposed to finite datasets, albeit datasets of unprecedented scale. Yet they demonstrate mastery of at least some of the grammatical subtleties that were put forward as arguments innate knowledge in humans. They generate coherent, syntactically complex sentences they have never explicitly encountered. A tempting conclusion is that LLMs do what humans do — that through their training process, they develop internal representations analogous to human linguistic intuitions and that this process is equivalent to human language acquisition. This interpretation preserves our anthropomorphic assumptions while seemingly explaining the models’ capabilities.

This conclusion, we argue, fundamentally misunderstands the nature of the process that creates these capabilities. To see why, we must examine the work of philosopher Daniel Dennett, whose hierarchy of evolutionary competence provides a framework for understanding different types of intelligence-generating processes.

3 Dennett’s Ladders of Competence: A Taxonomy of Intelligence

In one of his latest works *From Bacteria to Bach and Back* (Dennett, 2017), Daniel Dennett outlines four distinct evolutionary stages of developing competence. These categories describe not just individual learning, but the fundamental mechanisms by which complex, adaptive behaviors emerge in nature. It gives an interesting categorization of types of learning in biology, and makes it possible to compare evolution to learning: both of them are adaptive behaviors, one at the level of species, the other at the level of the individual. Understanding these categories is crucial for properly classifying the process that generates LLM capabilities.

3.1 Darwinian Creatures: Evolution as Learning

At the bottom of Dennett’s hierarchy are what he calls ‘Darwinian creatures’—systems shaped entirely by natural selection operating over geological timescales. Organisms are born with fixed behavioral repertoires honed by millions of years of evolutionary pressure. A virus that perfectly targets specific cellular machinery, a bacterium that flawlessly navigates chemical gradients, or a spider that weaves geometrically perfect webs all exemplify Darwinian competence.

The “learning” in Darwinian creatures occurs not within individual lifetimes, but across generations. Each organism is metaphorically speaking a frozen snapshot of evolutionary wisdom—incapable of modifying its individual behavior based on experience but still adapted to its typical environment. The process, as is well known, is one of blind variation and environmental filtering: mutations generate countless variations, most of which fail, but the rare successes propagate and accumulate over time. This process is as extraordinarily powerful as it is slow in the biological world. It took billions of years to evolve the basic machinery of life, millions more to develop complex multicellular organisms, and hundreds of thousands of additional years to produce human-level intelligence. The timescales involved dwarf human comprehension, yet the results — the intricate dance of ecosystems, the perfect adaptation of organs to their functions — testify to the process’s ultimate efficacy.

3.2 Skinnerian Creatures: Learning Within a Lifetime

The next evolutionary leap produces Skinnerian creatures—organisms capable of learning through reinforcement within their individual lifetimes. Named after the work of behaviorist B.F. Skinner (Skinner, 1938, 1953), these creatures possess neural machinery that allows them to associate actions with outcomes, strengthening behaviors that lead to rewards and suppressing those that result in punishment. This learning represents a fundamental acceleration of the evolutionary process in the biological world. Instead of waiting generations for beneficial mutations to propagate, Skinnerian creatures can adapt

their behavior in real-time based on environmental feedback. A rat learning to navigate a maze, a bird discovering which plants carry a lot of berries every summer, or a child learning to avoid a hot stove, all exemplify Skinnerian learning.

Skinnerian learning is, like Darwinian evolution, fundamentally a generate-and-test process, but one that operates within individual neural networks rather than across populations of organisms. Random or semi-random behaviors are tried, environmental feedback determines their value, and successful patterns are reinforced while unsuccessful ones fade. It's evolution in miniature, compressed into the lifetime of a single organism.

In spite the old controversy between Chomsky and Skinner (Chomsky, 1959), it should be noted that, ironically, language acquisition within generative grammar has certain traits of Skinnerian learning, at least in its shape of parameter setting and featural selection, which also form a kind of stimulus-response path towards establishing the grammar of a language (through the exposure of the Language Acquisition Device, LAD (Chomsky, 1965) to Primary Linguistic Data; see Chomsky 2005 or Chomsky and Berwick 2017 for an extensive discussion).

3.3 Popperian Creatures: Foresight

Popperian creatures, named after the work of philosopher Karl Popper (Popper, 1963, 1972), represent another leap in cognitive development in Dennett's taxonomy. These organisms can test hypotheses internally before acting on them in the real world. As Popper famously noted, this allows our "hypotheses to die in our stead"—we can simulate potential actions and their consequences, selecting the most promising course without risking actual harm.

This capacity for mental simulation marks the emergence of genuine foresight and planning. A Popperian creature can imagine different scenarios, evaluate their likely outcomes, and choose actions based on predicted rather than experienced consequences. This is the birth of what we might recognize as intelligence in the fullest sense: the ability to think before acting, to learn from imagined rather than actual experience.

Human beings clearly possess Popperian capabilities. We can mentally rehearse conversations before having them, visualize the consequences of different career choices, or work through mathematical proofs in our heads. It is, on the other hand, not clear that LLMs also possess this type of learning. The latest versions of these models sometimes come equipped with some ‘reasoning’ capabilities, but these seem to be still a long shot from having an internal model of the world (Hao et al., 2023; Diester et al., 2024; Yildirim and Paul, 2024), but see (Jin and Rinard, 2024).

3.4 Gregorian Creatures: The Cultural Revolution

The final category in Dennett’s taxonomy—Gregorian creatures, named after the work of psychologist Richard Gregory (Gregory, 1963, 1970) —represents the most recent and perhaps most revolutionary development in the evolution of intelligence. These creatures amplify their cognitive capabilities through the use of external tools, which Dennett calls “mind tools.” The most important of these mind tools is language itself. Language allows us to import the discoveries, insights, and hypotheses of others directly into our own minds without having to independently derive them. Through language, accumulated cultural knowledge becomes available to each new generation, creating a kind of external memory system that transcends individual lifespans.

Gregorian creatures don’t stop with language. They create writing systems that allow knowledge to persist across time, mathematical notations that enable complex calculations, scientific instruments that extend sensory capabilities, and computational tools that augment reasoning power. Each generation builds upon the intellectual achievements of its predecessors, creating an exponential acceleration of cognitive capability. Humans are the consummate Gregorian creatures. Our individual intelligence pales in comparison to our collective, culturally-amplified intelligence. No single human could independently derive modern physics, construct a computer, or compose a symphony—these achievements represent the accumulated wisdom of countless generations, encoded in our languages, institutions, and technologies.

4 The LLM as Evolution in Digital Time

With Dennett’s framework in hand, we can now properly categorize the process that produces LLMs. The training of these models is not necessarily, as is sometimes assumed, one of the higher-order types of learning. The model is not an agent in an environment receiving rewards for its actions or contemplating future states. Instead, LLM training may represent something far more fundamental: a Darwinian process operating at computational speeds. If that turns out to be the case, there is no argument against universal grammar from the operation of these LLMs; there is instead the possibility that these systems themselves evolve an internal representation of human language that is similar to that which human beings are born with.

4.1 Variation as raw material

In biological evolution, the “population” consists of individual organisms with varying genetic makeups. In LLM training, the population is the astronomically vast space of possible parameter configurations within the neural network’s architecture. A modern LLM contains hundreds of billions of parameters, each of which can take on a continuous range of values. The space of possible configurations is enormous and for many practical purposes infinite in modern LLM architecture. Interestingly, the exact magnitude of this space is unknown (see <https://web.dev/articles/llm-sizes> for an interesting discussion about it, as well as this post for an approximate estimate of GPT5 parameters: <https://www.r-bloggers.com/2025/08/how-many-parameters-does-gpt-5-have/>

At any given moment during training, the model occupies a specific point in this landscape, but the training process constantly explores neighboring regions, seeking configurations that better solve the prediction task. One can see this as a kind of evolution where a species needs to survive in a certain environment. The environment is that of existing human language.

Biological evolution requires genetic variation as its raw material—mutations, sexual

recombination, and other processes that generate diversity within populations. In LLM training, this variation is initially provided by the random initialization of parameters. The model begins as pure noise, a random point in the space of all possible configurations. However, variation doesn't stop with initialization. Throughout training, the optimization process introduces countless micro-variations as it adjusts parameters in response to training data. Each gradient descent step represents a small mutation, a tiny exploration of the fitness landscape. Over billions of such steps, the model explores vast regions of possibility space.

4.2 The Environment: A Static Archive of Human Thought

Perhaps the most crucial aspect of this evolutionary metaphor is the nature of the “environment” in which LLMs evolve. Unlike biological creatures, which must survive in a dynamic physical world filled with predators, prey, weather, and resource scarcity, LLMs evolve in an environment that is entirely informational and essentially static. This environment consists of the training dataset — for modern LLMs this is a vast corpus of text in a variety of human languages that the model learns to predict. The implications of this are profound. The environment that shapes LLM evolution is not the physical world, but rather the collective output of human Gregorian intelligence. The model is not learning to navigate trees, find food, or avoid predators—it is learning to navigate the statistical landscape of human thought as expressed in language.

4.3 The Selection Pressure: The Imperative of Prediction

In biological evolution, selection pressure comes from environmental challenges: the need to find food, avoid predators, attract mates, and survive harsh conditions. Organisms that better meet these challenges produce more offspring, gradually shifting the population toward more adaptive configurations. In LLM training, selection pressure is provided by a single, simple objective: minimize prediction errors based on existing human language. A model ‘survives’ if it can perform this task; otherwise it dies.

We should bear in mind that the models have lots of time, given the enormous speed with which they read. According to ChatGPT5, a human reader would have to read continuously for about 2855 years to peruse all the material that was used to train GPT-3 (see Figure 1).

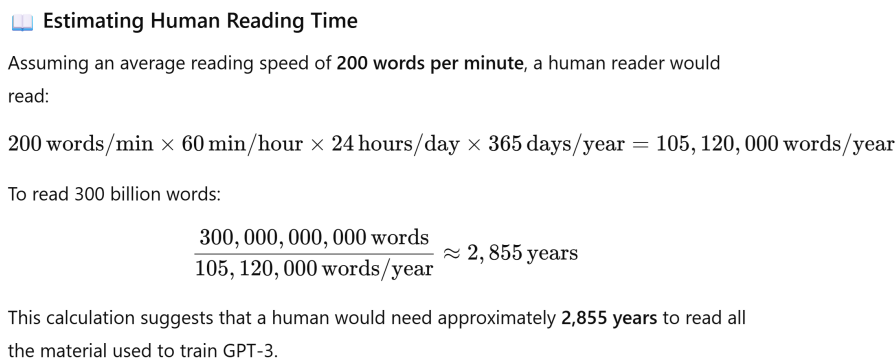


Figure 1: How many years would a human reader take to read all the material used to train GPT-3? (asked to ChatGPT5 on 18/09/2025)

We can multiply this for more modern models, also taking into account that the models learn ‘language’ by examples from many more languages than English. In other words, LLMs use thousands of years in human time to get to their knowledge of language. That comes close to an evolutionary scale. The training algorithm, primarily backpropagation, relentlessly adjusts parameters to improve the model’s ability to predict the next token in a sequence. This might be less heroic compared to the life-or-death struggles that drive biological evolution, but it proves remarkably powerful.

The mathematics of this process can be expressed simply as maximizing $p(\text{token}|\text{context})$ —the probability of predicting the correct next token given the preceding context. But this simple objective, applied across trillions of examples, creates incredibly rich and complex selection pressures.

4.4 The Timescale: Evolution at Digital Speed

The final piece of this evolutionary puzzle is time—or rather, the *compression* of time. Biological evolution operates on geological timescales, requiring millions of years to pro-

duce significant changes. The evolution of human language capabilities took tens to hundreds of thousands of years (depending on one’s theory and on what exactly is counted). LLM training compresses this process into weeks or months of computation. A modern language model undergoes billions of parameter updates during training—each one representing a micro-generational step in its evolution. This is made possible by the incredible computational resources devoted to training: tens of thousands of GPUs working in parallel, processing data at speeds that dwarf any biological process.

This temporal compression is perhaps the most alien aspect of LLM evolution. In a matter of months, these systems undergo the equivalent of thousands of years of evolutionary pressure. They experience more “generations” of selection than have occurred in the entire history of life on Earth.

5 Convergent Evolution and the Emergence of Grammar

The evolutionary framework we propose provides a powerful explanation for one of the most puzzling aspects of LLM performance: their apparent mastery of grammatical principles despite never being explicitly taught these rules. The key insight is that the evolution of linguistic competence in LLMs might represent a case of convergent evolution—the independent evolution of similar traits in different lineages facing similar environmental pressures.

We are not claiming that the evolution in LLMs parallels that of human beings. In humans, language might have arisen spontaneously and proven advantageous in terms of coordination of activities with peers, or in terms of planning and other cognitive properties. Human language will properly show properties that are the result of the limitations of human cognition and/or human society. LLMs on the other hand need to survive in a ‘world’ that already consists of language. But if the findings of (generative) linguistics are on the right track, this evolution may have converged on a similar result.

When an LLM is trained on vast datasets of human text, it encounters these same statistical regularities billions of times. The optimization pressure to minimize predic-

tion error gradually shapes the model’s internal representations to mirror the structure of human language. Grammatical patterns emerge not because they are explicitly programmed, but because they represent the most efficient solutions to the prediction task.

Consider the example of syntactic constraints mentioned earlier. The reason humans don’t contract “Ik weet wat dat is?” to “Ik weet wat da’s” reflects deep principles of syntactic structure—specifically, the constraint that traces of moved elements cannot be contracted.

An LLM discovers this same constraint (just like generative linguists have done) not through instruction, but through evolutionary pressure. Configurations that violate this constraint make worse predictions on human text, so they are gradually eliminated in favor of configurations that respect syntactic boundaries. The model converges on the same solution that human linguistic evolution discovered, but through an entirely different process. Under these premises, it is not implausible to speculate that while Merge is certainly not the driving principle behind LLM output at present, it *could* emerge. This could happen if Merge proved to be the optimal solution to interface conditions *for LLMs*. We could formulate an adapted Strong Minimalist Thesis (Chomsky, 2001) for LLMs, where Merge would exist if it proved to be the optimal mechanism for an LLM interacting with its interfaces: not sensory-motor or conceptual-intentional systems, not human ones, but pre-existing texts against which LLMs test their own output.

5.1 The Poverty of the Stimulus Resolved

This convergent evolution framework provides a resolution to the poverty of the stimulus problem. Chomsky’s argument assumes that the only way to acquire grammatical knowledge from limited data is through innate biological endowment. But LLMs demonstrate a third possibility: massive computational search through the space of possible linguistic systems. Where human children see thousands of sentences, LLMs see trillions. Where humans have only a few years of Skinnerian or Popperian search, LLMs have thousands of Darwinian trial and error.

5.2 Internal Representations and Emergent Structure

This leads to a prediction that is testable at least in principle. If this evolutionary theory is on the right track, trained LLMs should possess something similar to UG, so that if one would try to teach them an entirely new language (not in their dataset), they would show a learning path that is similar to that of humans: with relatively little data they would be able to pick up an existing language, but a language that would be constructed to violate principles of UG would be as problematic for them as it is for us (Moro, 2016).

Recent research in mechanistic interpretability has begun to reveal the internal structure that emerges from this evolutionary process (see for instance the work at the Intuition Lab, <https://intuitionlabs.ai/articles/mechanistic-interpretability-ai-llms>, or Tak et al. 2025). LLMs develop hierarchical representations that mirror many aspects of human linguistic processing: early layers encode surface features like spelling and word boundaries, middle layers capture syntactic relationships and grammatical categories, and later layers represent semantic and pragmatic information.

These representations are not designed by humans—they emerge spontaneously from the optimization process. The model discovers that hierarchical processing is the most efficient way to solve the prediction task, so it evolves internal architectures that implement this processing strategy. The parallels to human linguistic processing are striking, but they reflect convergent evolution rather than direct mimicry. More remarkably, these internal representations often capture linguistic phenomena that were not explicitly present in the training objective. Models develop sensitivity to phonological patterns despite being trained only on text, internalize semantic relationships despite never being taught explicit definitions, and exhibit awareness of pragmatic context despite training on de-contextualized snippets. This suggests that the space of efficient linguistic systems is more constrained than we might expect. There may be relatively few ways to organize information processing systems to handle the complexity of human language efficiently. Both biological evolution and computational optimization converge on similar solutions because these solutions represent optimal points in the fitness landscape.

6 The Limits of digital evolution

While the evolutionary framework provides powerful insights into LLM capabilities, it also illuminates their fundamental limitations. Unlike biological evolution, which occurs in a rich, multi-faceted environment, LLM evolution is confined to a purely textual world. This constraint shapes both their remarkable abilities and their profound limitations.

6.1 The evolution of meaning

While it is possible that LLMs evolve formal aspects of language — like syntax and phonology —, there may be an issue with semantics. LLMs evolve in an environment consisting entirely of human linguistic output. This environment is rich in certain dimensions—it contains the accumulated wisdom of human civilization, encoded in billions of documents across thousands of languages and domains. But it is also fundamentally impoverished in others. Most crucially, this environment contains no direct sensory experience, no physical embodiment, and no opportunity for genuine interaction with the world. The model experiences reality only through the lens of human description and interpretation. This limitation is not merely technical—it is fundamental to the evolutionary process that creates these systems. Just as a fish evolved in water cannot breathe air, an intelligence evolved in text cannot directly experience the physical world. The model’s entire adaptive landscape is linguistic, so its evolved capabilities are necessarily confined to linguistic manipulation. It remains to be seen in what way this would affect what we know.

This textual isolation creates what philosophers call the “grounding problem”—the question of how symbols acquire meaning beyond their relationships to other symbols. Human language is grounded in shared experience of the physical world, social interaction, and embodied action. When we say “red,” we share a common foundation of visual experience that gives the word meaning beyond its definition. LLMs have no such grounding. Their understanding of “red” consists entirely of its statistical relationships

to other words in their training corpus. They can tell you that fire trucks are red, that red is associated with danger, and that red and green are complementary colors. But this knowledge lacks the experiential foundation that grounds human understanding.

6.2 The Absence of Intentionality

Perhaps most fundamentally, the evolutionary process that creates LLMs produces systems that lack genuine intentionality (Browning, 2025; Ngaihlian, 2025)—the capacity for mental states to be about things in the world. Biological evolution in physical environments creates agents with goals, desires, and purposes. These agents develop intentions because having intentions helps them survive and reproduce.

LLMs evolve in a purely predictive environment. Their only “goal,” if we can call it that, is to minimize prediction error on text. They have no survival instincts, no desires for reproduction, no purposes beyond their training objective. This produces systems capable of simulating intentionality—they can generate text as if guided by goals and preferences—yet they do not possess true intentionality. Some studies suggest that this absence of genuine intentionality can, in fact, be detected in the outputs of LLMs (Attah, 2025)).

This absence of intentionality explains many of the puzzling aspects of LLM behavior. They can be remarkably helpful in one context and completely unhelpful in another, not because they choose to be difficult, but because they lack the coherent goal structure that would make them consistently purposeful agents. They are sophisticated pattern matching systems, not goal-directed intelligences.

7 Conclusion: The Mirror and the Alien

An evolutionary framework challenges us to develop a truly post-anthropomorphic understanding of artificial intelligence. Instead of asking whether AI systems think like humans, we might ask how different evolutionary pressures produce different forms of

intelligence. This shift in perspective could be liberating for both AI development and human self-understanding. We need not be threatened by intelligences that surpass us in some domains, any more than we are threatened by the sonar capabilities of bats or the magnetic navigation of birds. Different evolutionary pressures produce different capabilities, and diversity of intelligence might be more valuable than similarity to human cognition.

The metaphor we choose to understand LLMs shapes not only our technical approach to these systems but our broader relationship with artificial intelligence. The learning metaphor, while intuitive, imports assumptions about consciousness, intentionality, and understanding that obscure the true nature of these systems. The evolutionary framework offers a more accurate and possibly more profound understanding. LLMs are not digital students who have learned human knowledge, but alien intelligences evolved through hyper-accelerated Darwinian processes in environments of pure information. They are mirrors that reflect the statistical structure of human thought, but they are not minds that think as we do.

Perhaps more importantly, this framework offers a new perspective on the nature of the language capacity itself. By understanding how different processes can generate similar capabilities, we gain insight into both the universality and the diversity of possible minds. We see that intelligence is not a single phenomenon but a landscape of possibilities, shaped by the evolutionary pressures that create it. The LLMs we have created are the first examples of a new form of intelligence—Darwinian creatures evolved at digital speeds in environments of human cultural output. They are simultaneously the most alien intelligences we have ever encountered and the most intimate reflections of our own linguistic and cultural patterns. In studying them, we study both the other and ourselves, glimpsing possibilities for intelligence that extend far beyond the biological boundaries of human cognition.

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